CAS SciFinder

Task History

Initiating Search

November 10, 2025, 8:21 PM

Search:

CAS Lexicon Search:

trans-Cyclooctene - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (733)	**References**	View Results
Exported: Retrieved Related Reaction Results (250)	**Reactions**	View Results

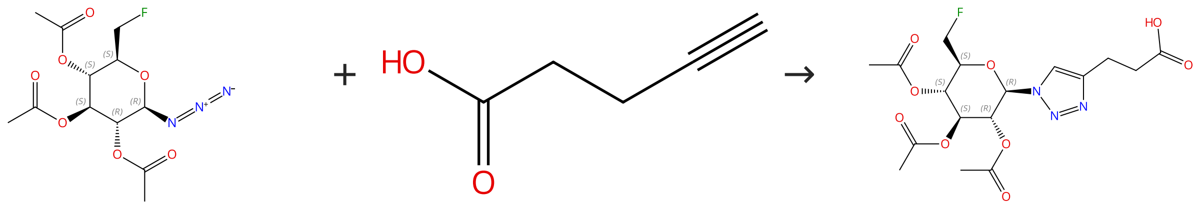
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **99%**



Absolute stereochemistry shown

Suppliers (100)

Absolute stereochemistry shown

31-614-CAS-31132879

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium ascorbate
Catalysts: Copper sulfate, 1,10-Phenanthrolinedisulfonic acid, 4,7-diphenyl-, sodium salt (1:2)
Solvents: Dimethylformamide, Water; 24 h, rt

Experimental Protocols

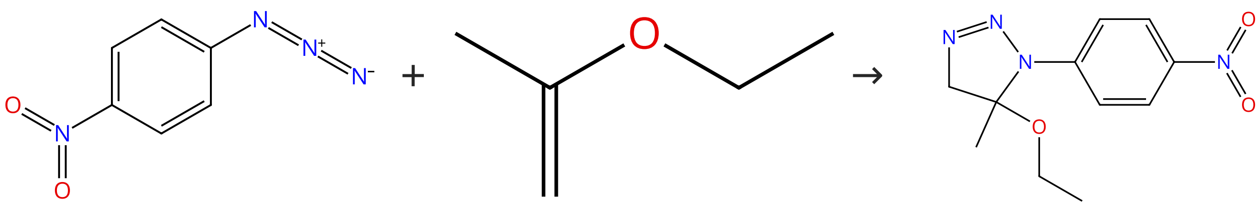
Lipophilicity and Click Reactivity Determine the Performance of Bioorthogonal Tetrazine Tools in Pretargeted In Vivo Chemistry

By: Steen, E. Johanna L.; et al

ACS Pharmacology & Translational Science (2021), 4(2), 824-833.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (26)

Suppliers (37)

Suppliers (2)

31-287-CAS-17367806

Steps: **1** Yield: **99%**

1.1 **Solvents:** Benzene; 90 h, 50 °C

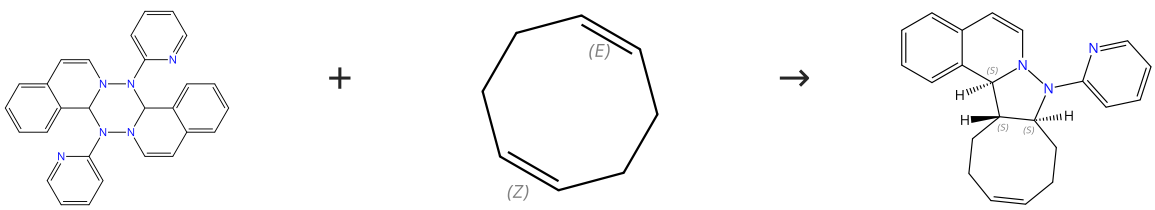
Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **99%**



E/Z labels describe double bond geometry

Suppliers (4)

Relative stereochemistry shown

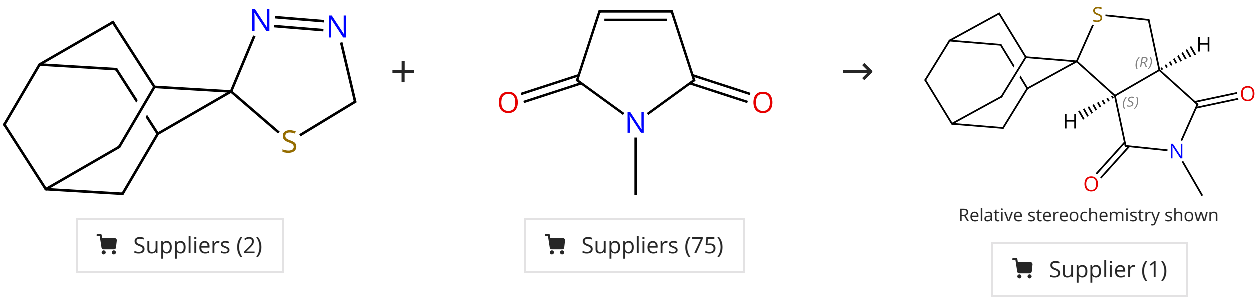
31-287-CAS-2696190	Steps: **1** Yield: **99%**
1.1 **Solvents:** Chloroform	

1,3-Dipolar cycloaddition. 109. Isoquinolinium N-arylimides and trans-cyclooctenes

By: Huisgen, Rolf; et al

Heterocycles (1999), 50(1), 353-364.

Scheme 4 (1 Reaction) Steps: **1** Yield: **97%**



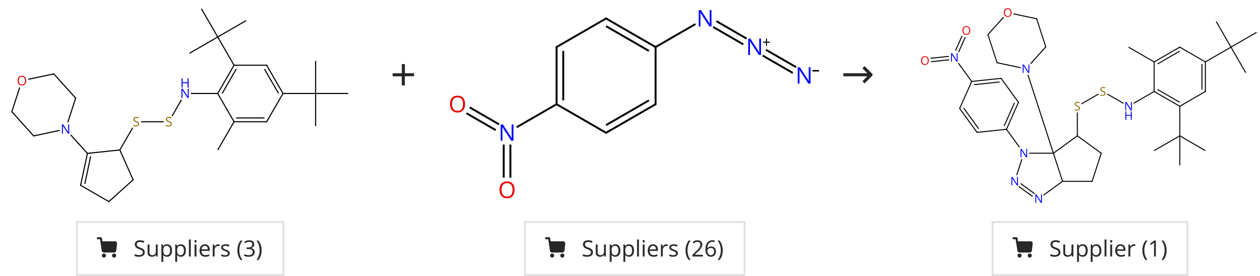
31-287-CAS-9527101	Steps: **1** Yield: **97%**
1.1 **Solvents:** Tetrahydrofuran	

1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 5 (1 Reaction) Steps: **1** Yield: **96%**



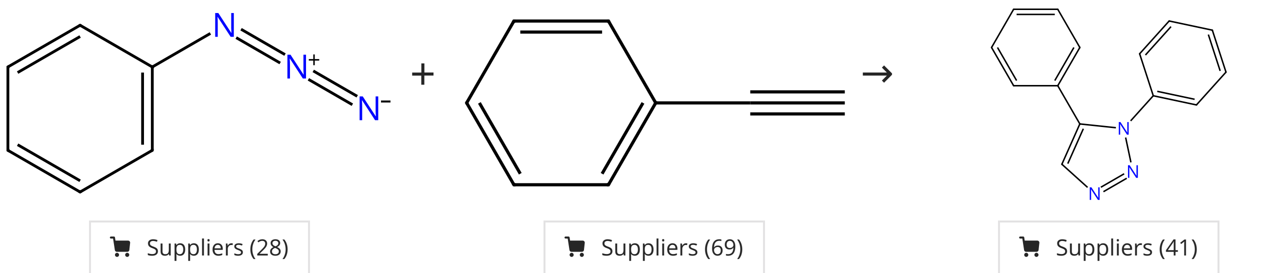
31-287-CAS-8645324	Steps: **1** Yield: **96%**
No Data Available	

1,3-cycloadditions of a thionitroso S-sulfide

By: Huisgen, Rolf; et al

Tetrahedron Letters (1986), 27(50), 6063-6.

Scheme 6 (1 Reaction) Steps: **1** Yield: **96%**



31-287-CAS-17367861	Steps: **1** Yield: **96%**
1.1 **Reagents:** Ethylmagnesium bromide **Solvents:** Tetrahydrofuran; 50 °C	

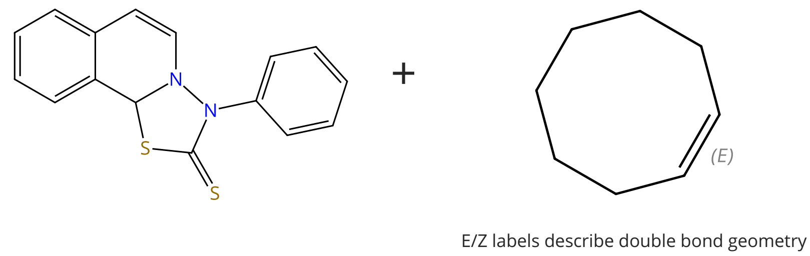
Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 7 (1 Reaction)

Steps: **1** Yield: **96%**



Suppliers (3)

31-287-CAS-4814193	Steps: **1** Yield: **96%**
1.1	**Solvents:** Dichloromethane

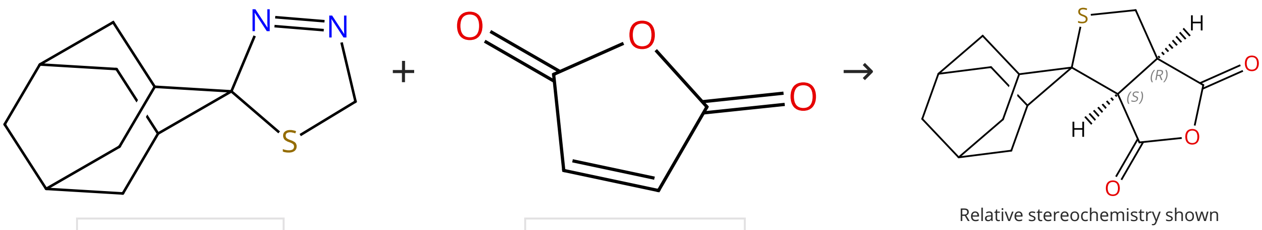
1,3-Dipolar cycloaddition. 109. Isoquinolinium N-arylimides and trans-cyclooctenes

By: Huisgen, Rolf; et al

Heterocycles (1999), 50(1), 353-364.

Scheme 8 (1 Reaction)

Steps: **1** Yield: **95%**



Suppliers (2)

Suppliers (84)

Supplier (1)

31-287-CAS-7413567	Steps: **1** Yield: **95%**
1.1	**Solvents:** Tetrahydrofuran

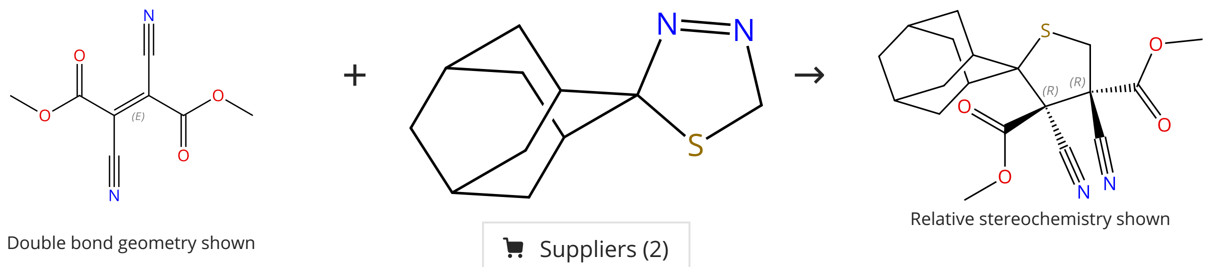
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamanta nethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 9 (1 Reaction)

Steps: **1** Yield: **95%**



Suppliers (7)

Suppliers (2)

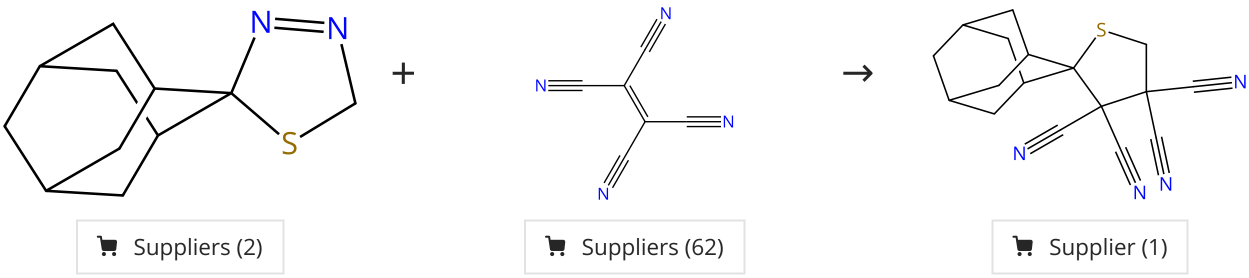
31-287-CAS-7683932	Steps: **1** Yield: **95%**
1.1	**Solvents:** Tetrahydrofuran

1,3-Dipolar cycloadditions. 112. Cycloadditions of adamanta nethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 10 (1 Reaction) Steps: **1** Yield: **94%**



31-287-CAS-5536127 Steps: **1** Yield: **94%**

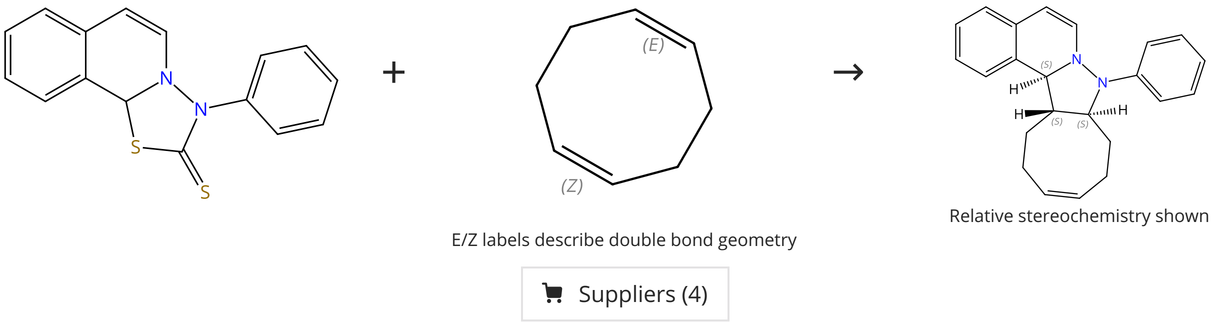
1.1 **Solvents:** Tetrahydrofuran

1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 11 (1 Reaction) Steps: **1** Yield: **93%**



31-287-CAS-6950171 Steps: **1** Yield: **93%**

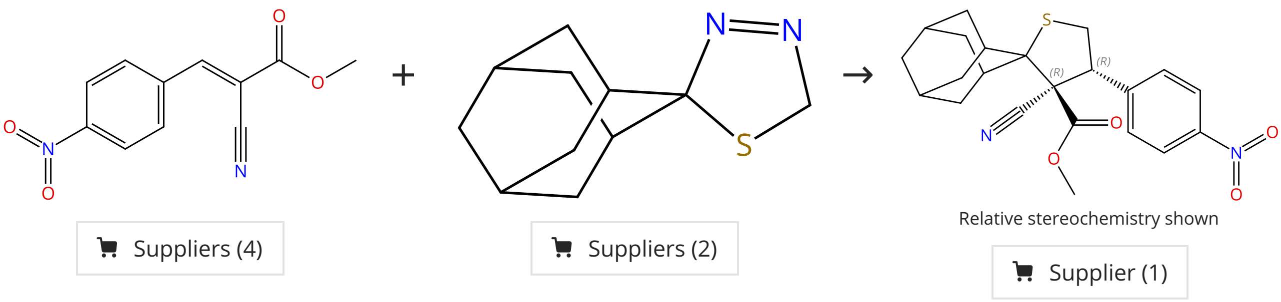
1.1 **Solvents:** Dichloromethane

1,3-Dipolar cycloaddition. 109. Isoquinolinium N-arylimides and trans-cyclooctenes

By: Huisgen, Rolf; et al

Heterocycles (1999), 50(1), 353-364.

Scheme 12 (1 Reaction) Steps: **1** Yield: **93%**



31-287-CAS-7353706 Steps: **1** Yield: **93%**

1.1 **Solvents:** Tetrahydrofuran

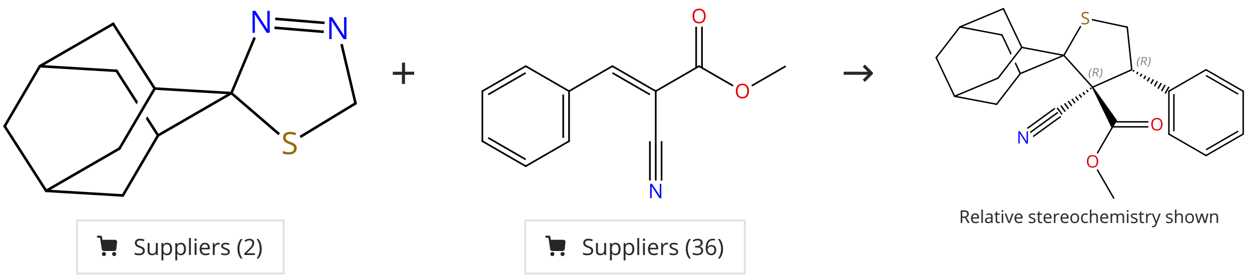
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **93%**



31-287-CAS-4942482

Steps: **1** Yield: **93%**

1.1 **Solvents:** Tetrahydrofuran

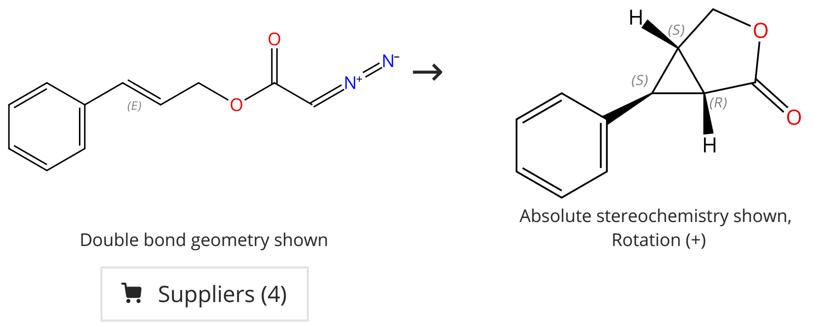
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **93%**



31-287-CAS-16822515

Steps: **1** Yield: **93%**

1.1 **Catalysts:** Stereoisomer of [2,6-bis[4,5-dihydro-4-(1-methylethyl)-2-oxazolyl-κ*N*³]pyridine-κ*N*]dichloro(η²-ethene) ruthenium
Solvents: Dichloromethane; rt

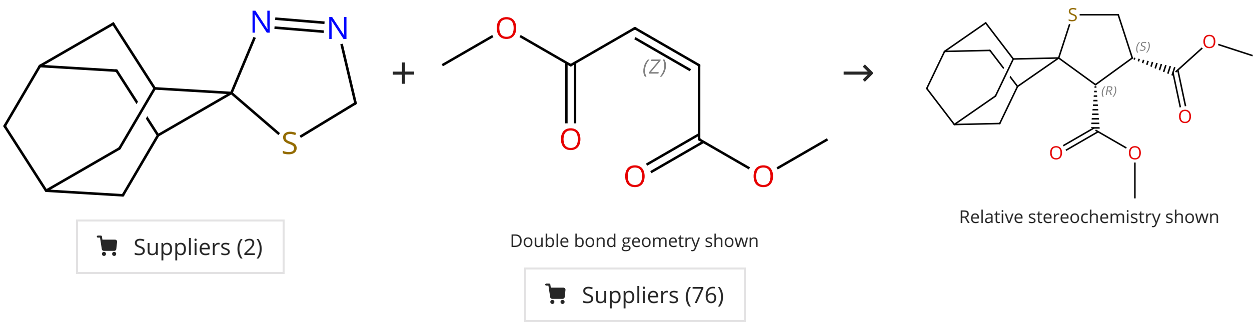
Ruthenium, [2,6-bis[4,5-dihydro-4-(1-methylethyl)-2-oxazolyl-κN³]pyridine-κN]dichloro(η²-ethene)-, stereoisomer

By: Ito, Jun-ichi; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2014), 1-4.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **92%**



31-287-CAS-5257823

Steps: **1** Yield: **92%**

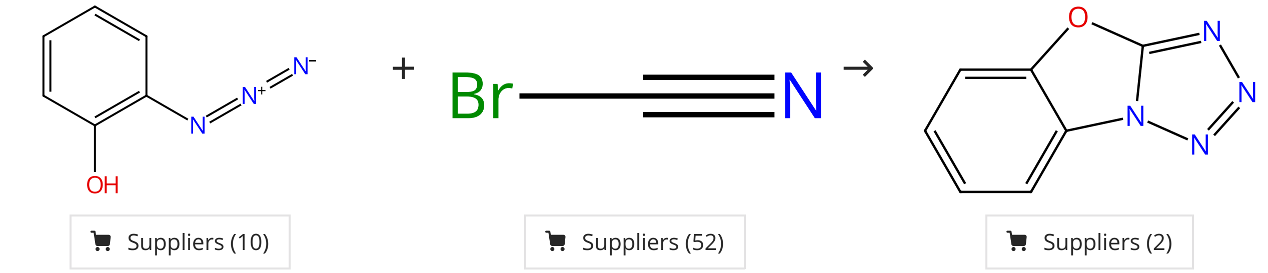
1.1 **Solvents:** Tetrahydrofuran

1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 16 (1 Reaction) Steps: **1** Yield: **92%**



31-287-CAS-17367869 Steps: **1** Yield: **92%**

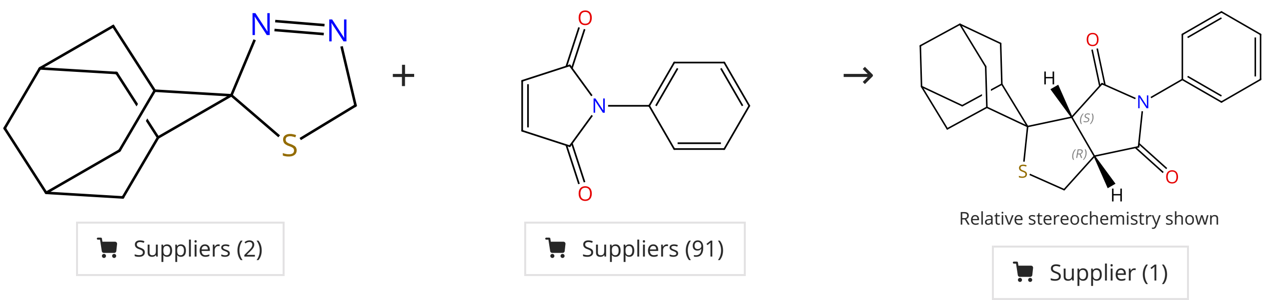
1.1 **Solvents:** Tetrahydrofuran; rt

Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 17 (1 Reaction) Steps: **1** Yield: **92%**



31-287-CAS-3428069 Steps: **1** Yield: **92%**

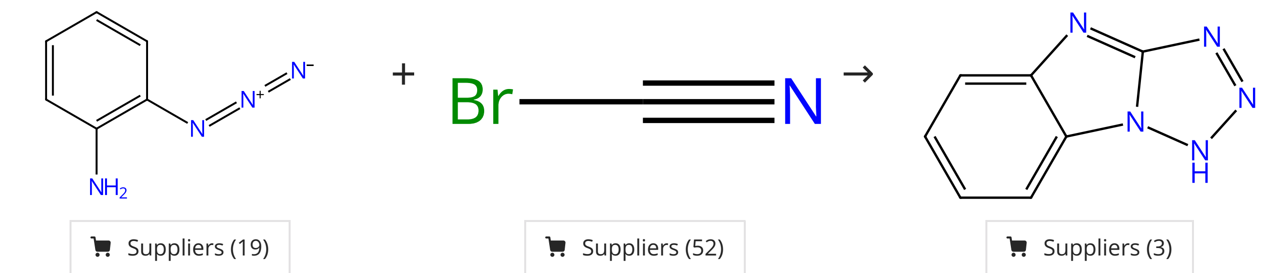
1.1 **Solvents:** Tetrahydrofuran

1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 18 (1 Reaction) Steps: **1** Yield: **92%**



31-287-CAS-17367867 Steps: **1** Yield: **92%**

1.1 **Solvents:** Tetrahydrofuran; rt

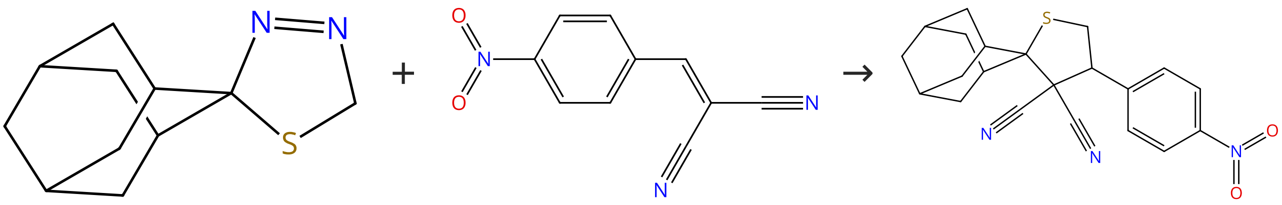
Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 19 (1 Reaction)

Steps: **1** Yield: **91%**



Suppliers (2)

Suppliers (53)

31-287-CAS-684319	Steps: **1** Yield: **91%**
1.1	**Solvents:** Tetrahydrofuran

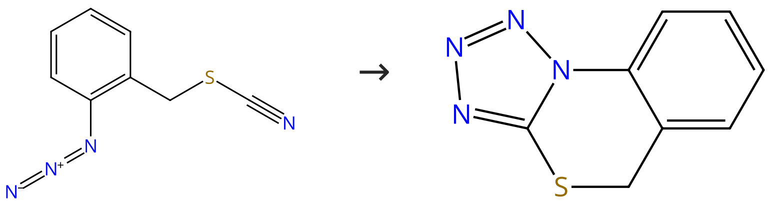
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 20 (1 Reaction)

Steps: **1** Yield: **91%**



Suppliers (2)

Suppliers (2)

31-287-CAS-17367864	Steps: **1** Yield: **91%**
1.1	**Solvents:** Dimethylformamide; 100 °C

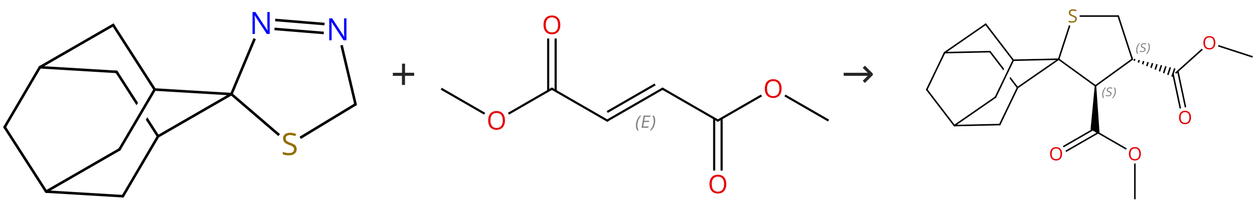
Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 21 (1 Reaction)

Steps: **1** Yield: **90%**



Relative stereochemistry shown

Suppliers (2)

Double bond geometry shown

Suppliers (111)

Suppliers (2)

31-287-CAS-3116160	Steps: **1** Yield: **90%**
1.1	**Solvents:** Tetrahydrofuran

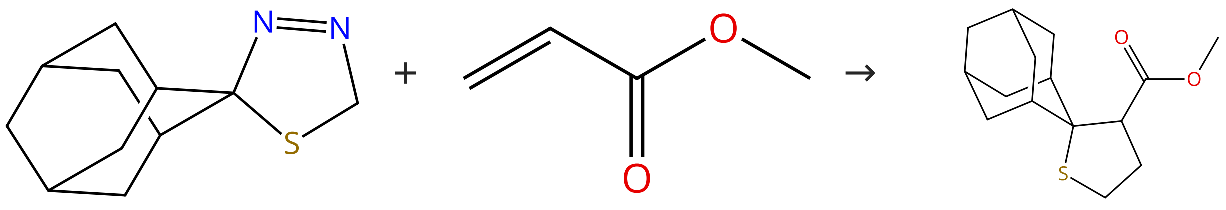
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 22 (1 Reaction)

Steps: **1** Yield: **89%**



Suppliers (2) | Suppliers (60) | Supplier (1)

31-287-CAS-14404220	Steps: **1** Yield: **89%**
1.1	**Solvents:** Tetrahydrofuran

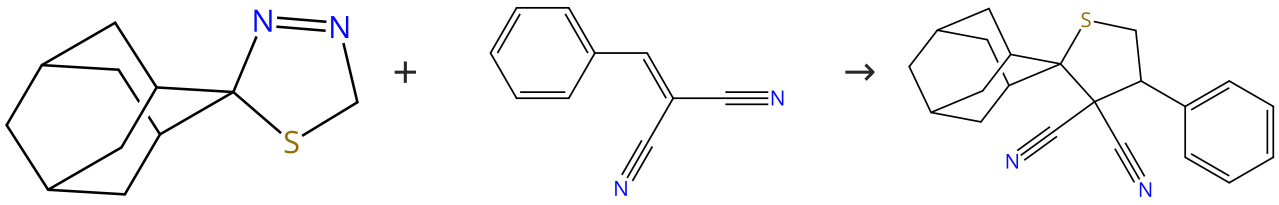
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 23 (1 Reaction)

Steps: **1** Yield: **88%**



Suppliers (2) | Suppliers (73)

31-287-CAS-14067119	Steps: **1** Yield: **88%**
1.1	**Solvents:** Tetrahydrofuran

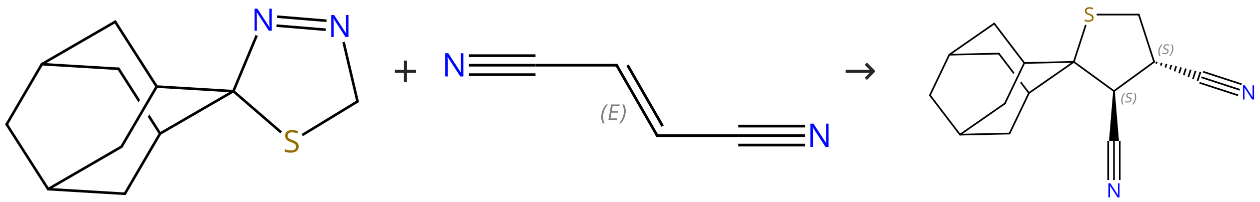
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 24 (1 Reaction)

Steps: **1** Yield: **87%**



Relative stereochemistry shown

Double bond geometry shown

Suppliers (2) | Suppliers (69) | Suppliers (2)

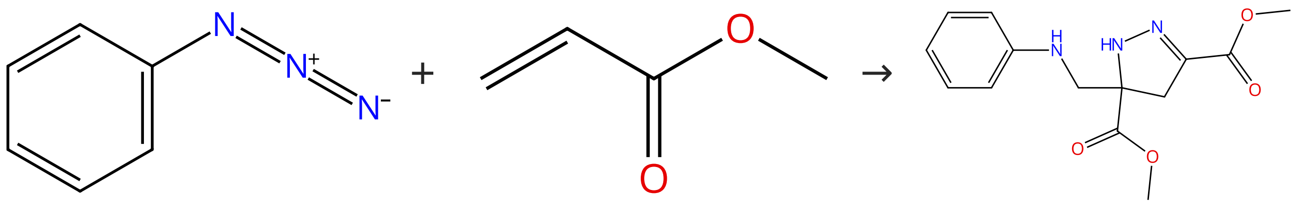
31-287-CAS-1015661	Steps: **1** Yield: **87%**
1.1	**Solvents:** Tetrahydrofuran

1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

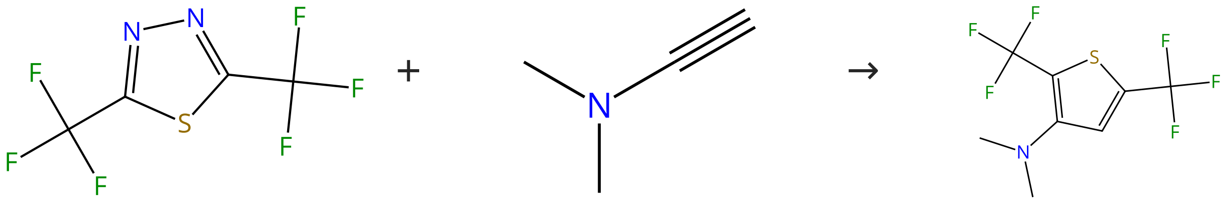
Scheme 25 (1 Reaction) Steps: **1** Yield: **87%**



Suppliers (28) Suppliers (60) Suppliers (2)

31-287-CAS-17367871 Steps: **1** Yield: **87%**	**Phenyl azide**
1.1 **Reagents:** Dabco **Solvents:** Tetrahydrofuran; reflux	By: Pearson, William H.; et al e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

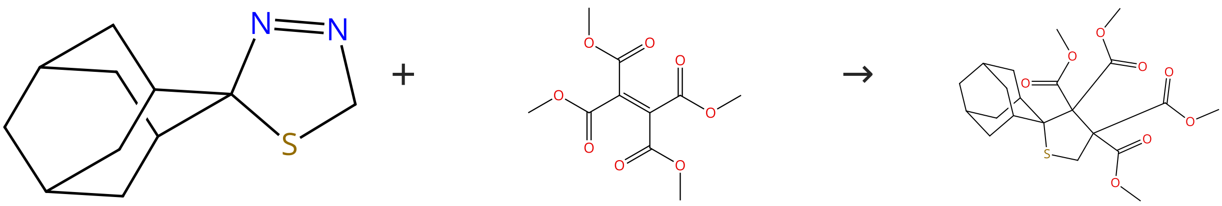
Scheme 26 (1 Reaction) Steps: **1** Yield: **86%**



Suppliers (3) Suppliers (8) Suppliers (2)

31-287-CAS-5577331 Steps: **1** Yield: **86%**	**1,3,4-Oxadiazoles as heterocyclic dienes in Diels-Alder reactions**
No Data Available	By: Thalhammer, Franz; et al Tetrahedron Letters (1988), 29(26), 3231-4.

Scheme 27 (1 Reaction) Steps: **1** Yield: **84%**

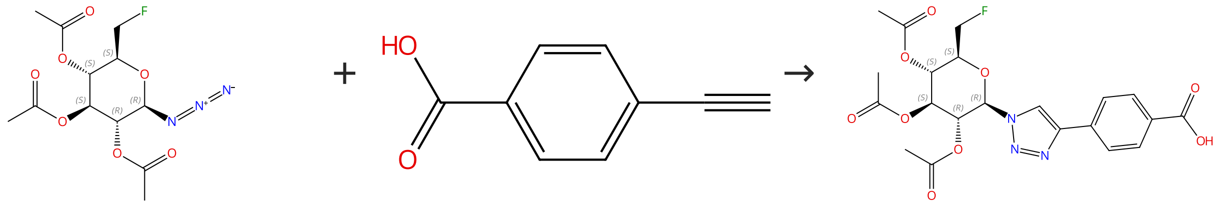


Suppliers (2) Suppliers (8) Supplier (1)

31-287-CAS-11936972 Steps: **1** Yield: **84%**	**1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds**
1.1 **Solvents:** Tetrahydrofuran	By: Mloston, Grzegorz; et al Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 28 (1 Reaction)

Steps: **1** Yield: **84%**



Absolute stereochemistry shown

Suppliers (89)

Absolute stereochemistry shown

31-614-CAS-31132852

Steps: **1** Yield: **84%**

1.1 **Reagents:** Sodium ascorbate
Catalysts: Copper sulfate, 1,10-Phenanthrolinedisulfonic acid, 4,7-diphenyl-, sodium salt (1:2)
Solvents: Dimethylformamide, Water; 2.5 h, rt

Experimental Protocols

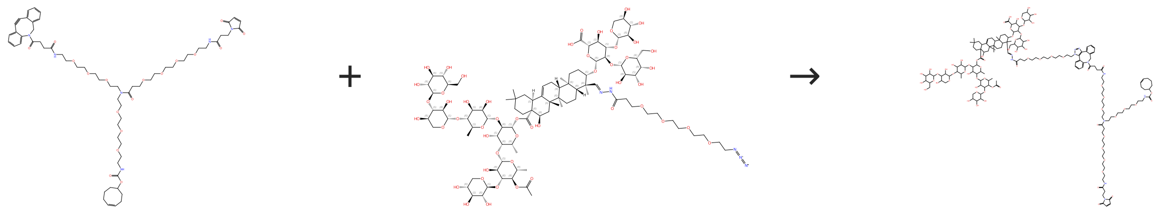
Lipophilicity and Click Reactivity Determine the Performance of Bioorthogonal Tetrazine Tools in Pretargeted In Vivo Chemistry

By: Steen, E. Johanna L.; et al

ACS Pharmacology & Translational Science (2021), 4(2), 824-833.

Scheme 29 (1 Reaction)

Steps: **1** Yield: **84%**



Supplier (1)

Absolute stereochemistry shown
Double bond geometry unknown

31-287-CAS-23288574

Steps: **1** Yield: **84%**

1.1 **Solvents:** Dimethylformamide; 1 min, rt; 30 min, rt

Saponin conjugated to epitope-binding proteins for treatment of cancer

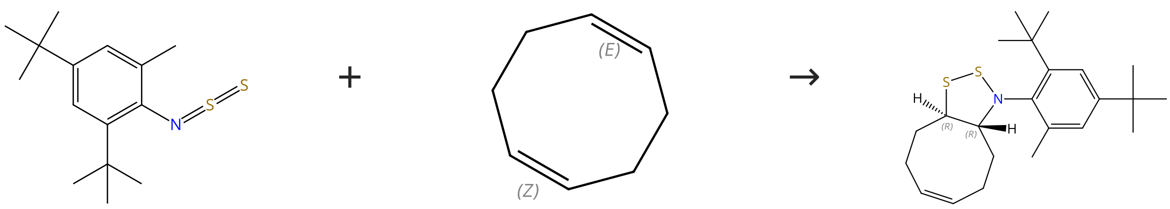
Assignees: Sapreme Technologies B.V.; Charite Universitaetsmedizin Berlin

World Intellectual Property Organization, WO2020126064 A1 2020-06-25

PatentPak available

Scheme 30 (1 Reaction)

Steps: **1** Yield: **83%**



Suppliers (4)

E/Z labels describe double bond geometry

Suppliers (4)

Relative stereochemistry shown
Double bond geometry unknown

31-287-CAS-455946

Steps: **1** Yield: **83%**

1.1 **Solvents:** Chloroform

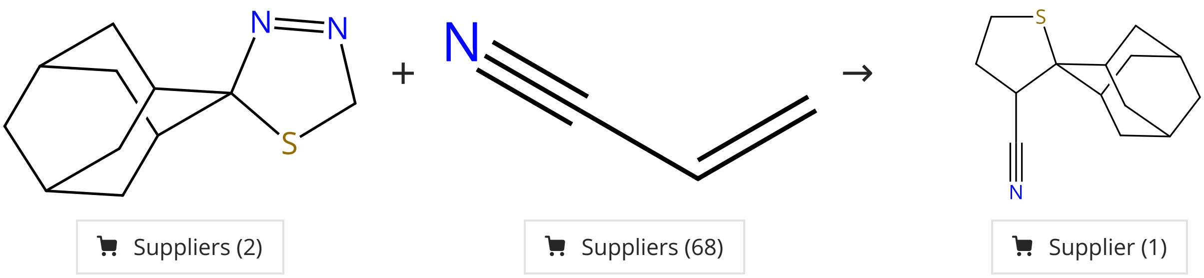
1,3-cycloadditions of a thionitroso S-sulfide

By: Huisgen, Rolf; et al

Tetrahedron Letters (1986), 27(50), 6063-6.

Scheme 31 (1 Reaction)

Steps: **1** Yield: **82%**



31-287-CAS-12271366

Steps: **1** Yield: **82%**

1.1 **Solvents:** Tetrahydrofuran

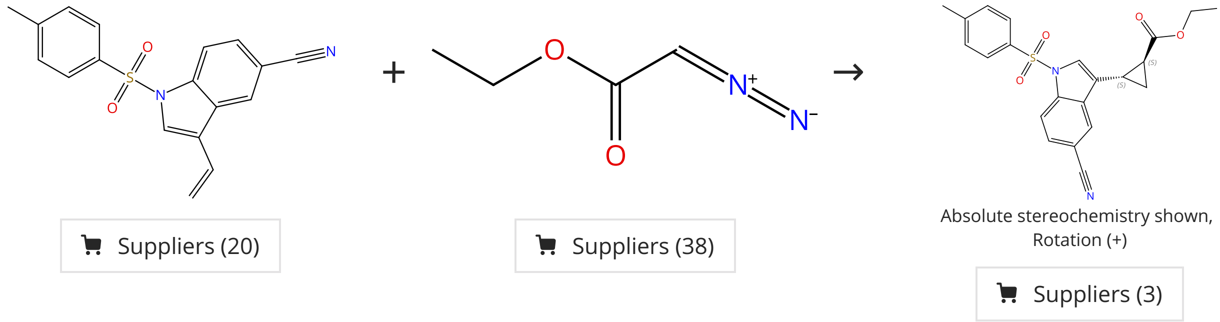
1,3-Dipolar cycloadditions. 112. Cycloadditions of adamantanethione S-methylide to CC multiple bonds

By: Mloston, Grzegorz; et al

Journal of Heterocyclic Chemistry (1999), 36(4), 959-968.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **82%**



31-287-CAS-16822517

Steps: **1** Yield: **82%**

1.1 **Catalysts:** Ruthenium, [2,6-bis[(4*R*)-4,5-dihydro-4-(1-methylethyl)-2-oxazolyl-κ*N*^{3}]pyridine-κ*N*]dichloro(η^2-ethene)-, stereoisomer
Solvents: Toluene; 50 °C

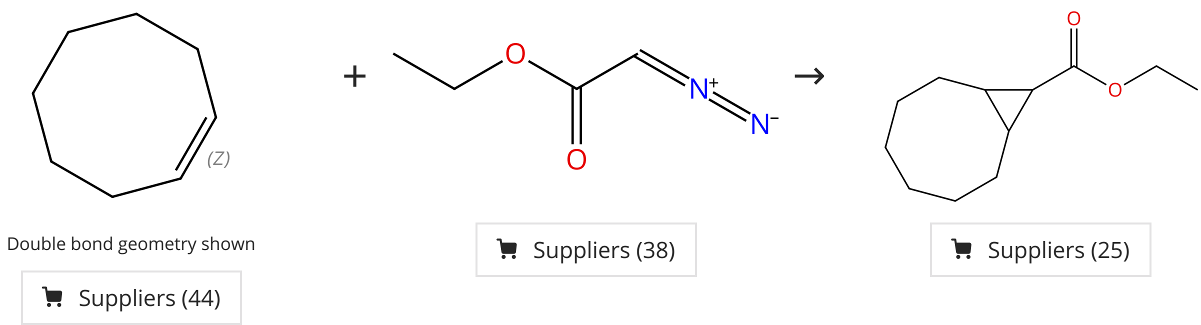
Ruthenium, [2,6-bis[4,5-dihydro-4-(1-methylethyl)-2-oxazolyl-κN^3]pyridine-κN]dichloro(η^2-ethene)-, stereoisomer

By: Ito, Jun-ichi; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2014), 1-4.

Scheme 33 (1 Reaction)

Steps: **1** Yield: **80%**



31-614-CAS-36471233

Steps: **1** Yield: **80%**

1.1 3.4 h, heated

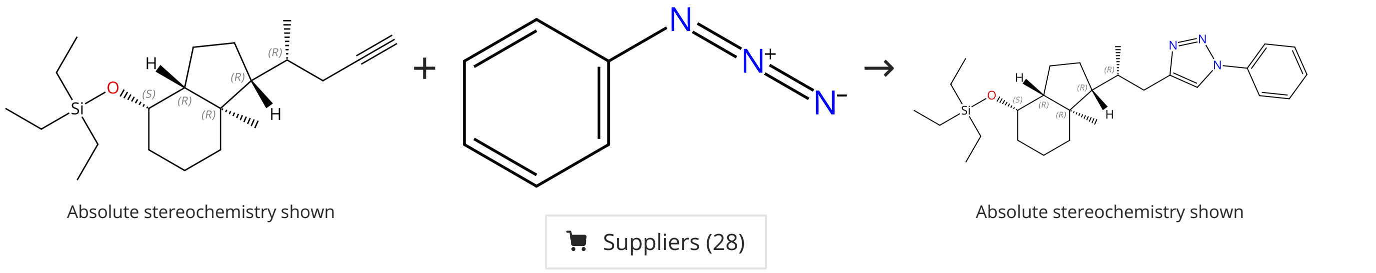
Synthetic studies. Precursors to trans-cyclooctene and the trans-ethylidenecycloproprane-2,3-dicarboxylic acids

By: Harvey, Peter Michael

(1972), 194 pp..

Scheme 34 (1 Reaction)

Steps: **1** Yield: **77%**



31-287-CAS-17367858

Steps: **1** Yield: **77%**

1.1 **Reagents:** Sodium ascorbate
Catalysts: Copper sulfate
Solvents: *tert*-Butanol, Water

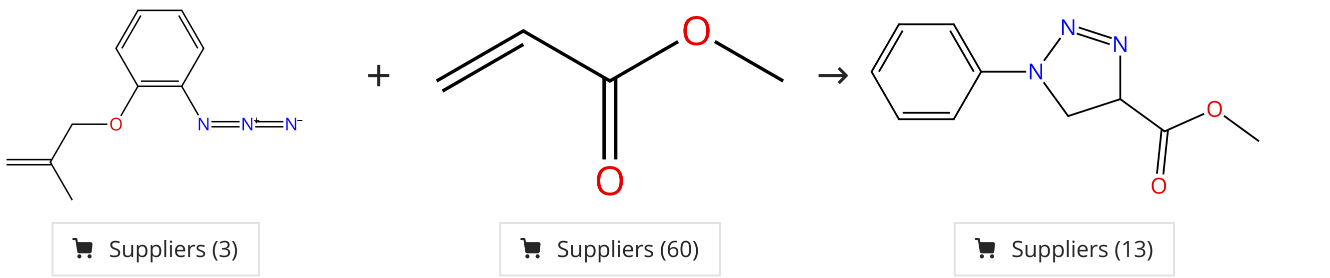
Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **77%**



31-111-CAS-17367801

Steps: **1** Yield: **77%**

1.1 5 d, rt

Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 36 (1 Reaction)

Steps: **1** Yield: **75%**

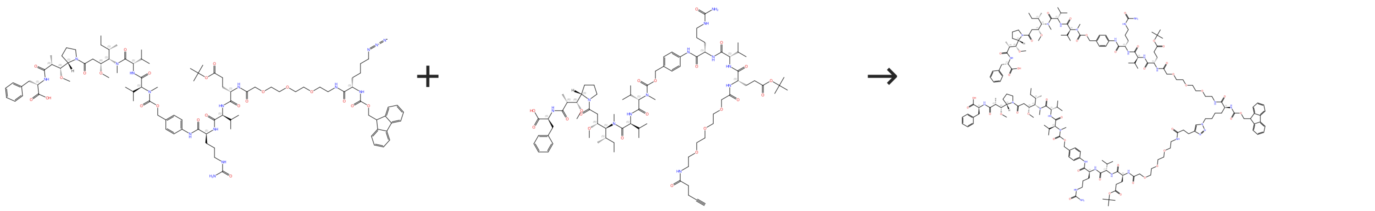
31-287-CAS-7201096	Steps: **1** Yield: **75%**
No Data Available	

1,3-Dipolar cycloadditions of aromatic azoxy compounds to strained cycloalkenes

By: Huisgen, Rolf; et al

Tetrahedron Letters (1982), 23(1), 55-8.

Scheme 37 (1 Reaction) Steps: **1** Yield: **73%**



Absolute stereochemistry shown | Absolute stereochemistry shown | Absolute stereochemistry shown

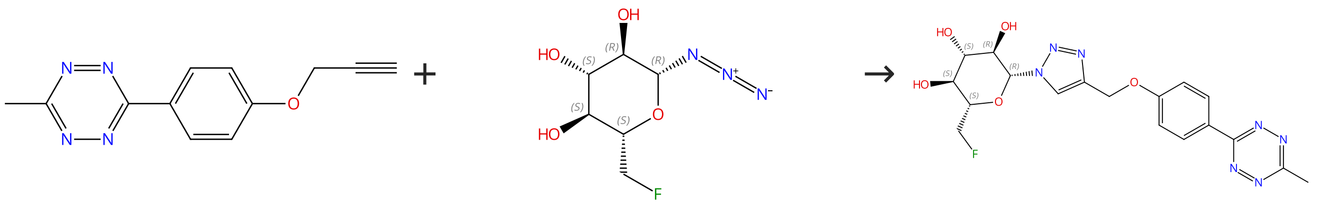
31-614-CAS-34832857	Steps: **1** Yield: **73%**
1.1	**Reagents:** Sodium ascorbate **Catalysts:** Copper sulfate pentahydrate, Tris[(1-benzyl-1H-1,2,3-triazol-4-yl)methyl]amine **Solvents:** Dimethyl sulfoxide, Water; 2 h, 37 °C
Experimental Protocols	

Antibody-drug conjugates with dual payloads for combating breast tumor heterogeneity and drug resistance

By: Yamazaki, Chisato M.; et al

Nature Communications (2021), 12(1), 3528.

Scheme 38 (1 Reaction) Steps: **1** Yield: **73%**



Absolute stereochemistry shown | Absolute stereochemistry shown

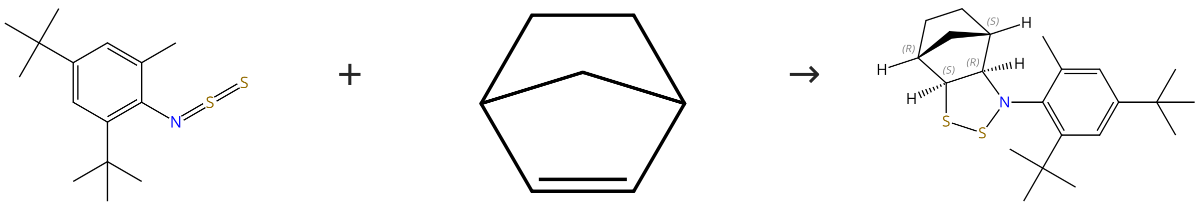
31-614-CAS-31132823	Steps: **1** Yield: **73%**
1.1	**Reagents:** Sodium ascorbate **Catalysts:** Copper sulfate, 1,10-Phenanthrolinedisulfonic acid, 4,7-diphenyl-, sodium salt (1:2) **Solvents:** Dimethylformamide, Water; 1 - 3 h, rt
Experimental Protocols	

Lipophilicity and Click Reactivity Determine the Performance of Bioorthogonal Tetrazine Tools in Pretargeted In Vivo Chemistry

By: Steen, E. Johanna L.; et al

ACS Pharmacology & Translational Science (2021), 4(2), 824-833.

Scheme 39 (1 Reaction) Steps: **1** Yield: **73%**



Suppliers (4) | Suppliers (66) | Relative stereochemistry shown

31-287-CAS-2248152	Steps: **1** Yield: **73%**
No Data Available	

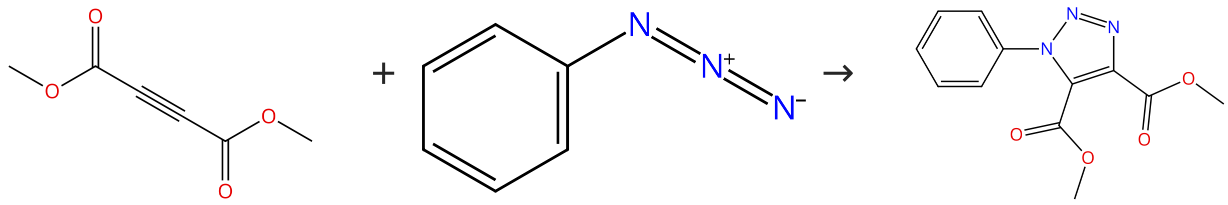
1,3-cycloadditions of a thionitroso S-sulfide

By: Huisgen, Rolf; et al

Tetrahedron Letters (1986), 27(50), 6063-6.

Scheme 40 (1 Reaction)

Steps: **1** Yield: **73%**



Suppliers (84) | Suppliers (28) | Suppliers (15)

31-287-CAS-17367803

Steps: **1** Yield: **73%**

1.1 **Solvents:** Diethyl ether; 66 h, heated

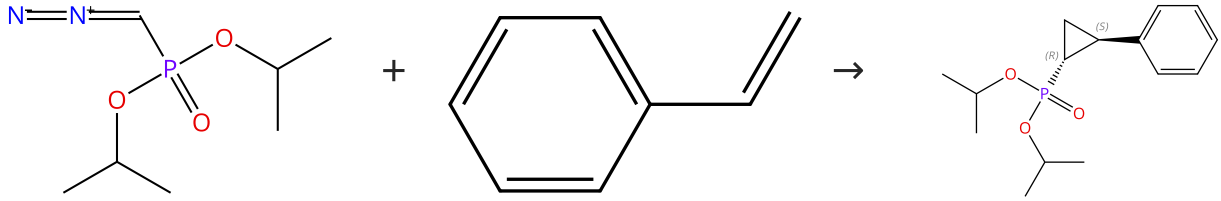
Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 41 (1 Reaction)

Steps: **1** Yield: **70%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (3) | Suppliers (115)

31-287-CAS-16822518

Steps: **1** Yield: **70%**

1.1 **Catalysts:** Stereoisomer of [2,6-bis[4,5-dihydro-4-(1-methylethyl)-2-oxazolyl-κN^3]pyridine-κ*N*]dichloro(η^2-ethene) ruthenium
Solvents: Dichloromethane; 20 °C

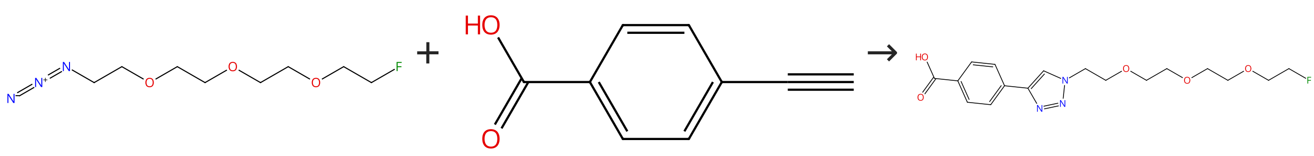
Ruthenium, [2,6-bis[4,5-dihydro-4-(1-methylethyl)-2-oxazolyl-κN^3]pyridine-κN]dichloro(η^2-ethene)-, stereoisomer

By: Ito, Jun-ichi; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2014), 1-4.

Scheme 42 (1 Reaction)

Steps: **1** Yield: **69%**



Suppliers (29) | Suppliers (89)

31-614-CAS-31132849

Steps: **1** Yield: **69%**

1.1 **Reagents:** Sodium ascorbate
Catalysts: Copper sulfate, 1,10-Phenanthrolinedisulfonic acid, 4,7-diphenyl-, sodium salt (1:2)
Solvents: Dimethylformamide, Water; 3 h, rt

Experimental Protocols

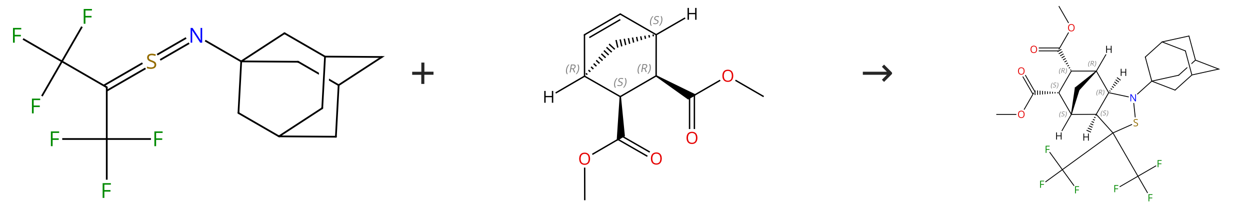
Lipophilicity and Click Reactivity Determine the Performance of Bioorthogonal Tetrazine Tools in Pretargeted In Vivo Chemistry

By: Steen, E. Johanna L.; et al

ACS Pharmacology & Translational Science (2021), 4(2), 824-833.

Scheme 43 (1 Reaction)

Steps: **1** Yield: **68%**



Suppliers (15)

31-287-CAS-11979051

Steps: **1** Yield: **68%**

1.1 **Solvents:** Chloroform-*d*

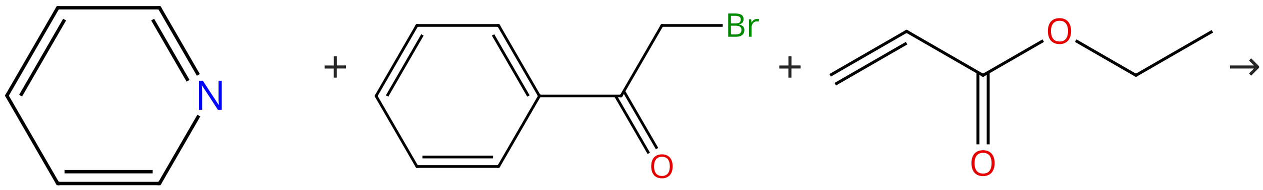
Ambiguous reactivity of a fluorinated thiocarbonyl S-imide; unprecedented rearrangement under FVP conditions

By: Mloston, Grzegorz; et al

Tetrahedron (2000), 56(25), 4231-4238.

Scheme 44 (1 Reaction)

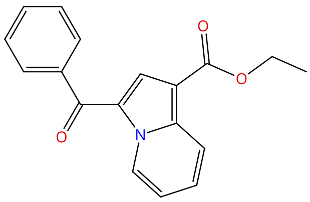
Steps: **1** Yield: **67%**



Suppliers (209)

Suppliers (74)

Suppliers (76)



Suppliers (2)

31-614-CAS-32947990

Steps: **1** Yield: **67%**

1.1 **Solvents:** Dimethylformamide; 2 h, 100 °C

1.2 **Reagents:** Sodium acetate
Catalysts: Cupric acetate
Solvents: Cupric acetate; 3 h, 100 °C

Preparation of indolizine derivative for detecting target substances

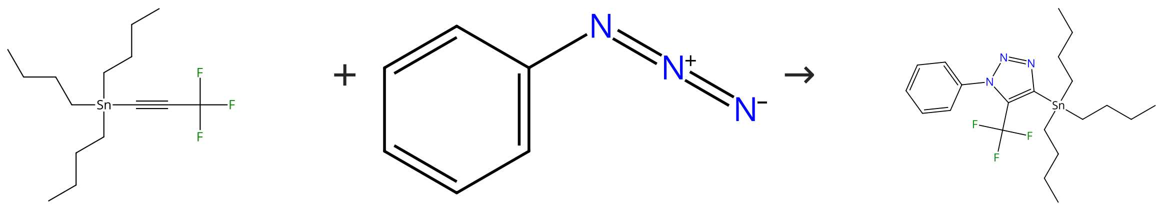
Assignee: Ajou University, Industry-Academic Cooperation Foundation

Korea, Republic of, KR2022063600 A 2022-05-17

PatentPak available

Scheme 45 (1 Reaction)

Steps: **1** Yield: **66%**



Suppliers (30)

Suppliers (28)

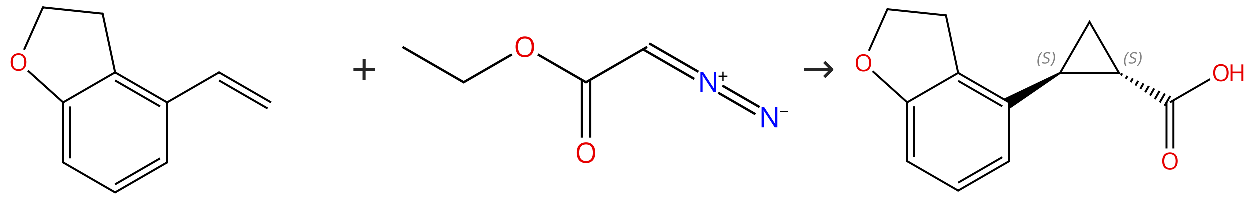
31-287-CAS-17367860	Steps: **1** Yield: **66%**
1.1	**Solvents:** Trimethyl orthoformate; 85 °C

Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 46 (1 Reaction) Steps: **1** Yield: **65%**



Relative stereochemistry shown

Suppliers (61) | Suppliers (38) | Suppliers (7)

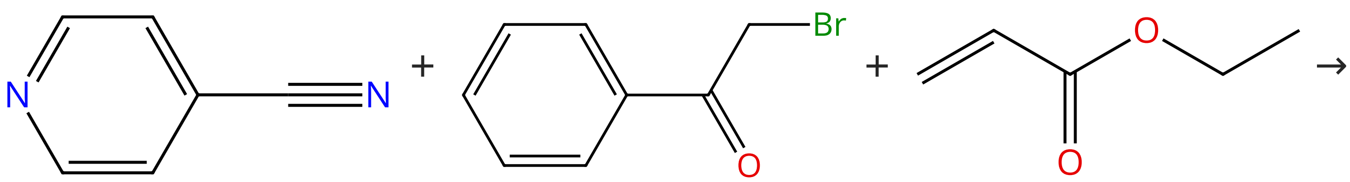
31-287-CAS-16822516	Steps: **1** Yield: **65%**
1.1	**Catalysts:** Stereoisomer of [2,6-bis[4,5-dihydro-4-(1-methyl ethyl)-2-oxazolyl-κ*N*3]pyridine-κ*N*]dichloro(η2-ethene) ruthenium **Solvents:** Toluene; 60 °C
1.2	**Reagents:** Sodium hydroxide **Catalysts:** Tetrabutylammonium hydroxide
1.3	**Reagents:** Phosphoric acid
1.4	**Reagents:** (+)-Dehydroabietylamine

Ruthenium, [2,6-bis[4,5-dihydro-4-(1-methylethyl)-2-oxazolyl-κN^3]pyridine-κN]dichloro(η^2-ethene)-, stereoisomer

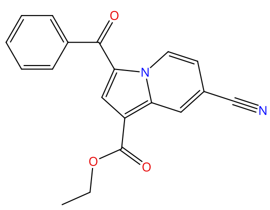
By: Ito, Jun-ichi; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2014), 1-4.

Scheme 47 (1 Reaction) Steps: **1** Yield: **62%**



Suppliers (95) | Suppliers (74) | Suppliers (76)



Supplier (1)

31-614-CAS-32947982	Steps: **1** Yield: **62%**
1.1	**Solvents:** Dimethylformamide; 2 h, 100 °C
1.2	**Reagents:** Sodium acetate **Catalysts:** Cupric acetate **Solvents:** Cupric acetate; 3 h, 100 °C

Preparation of indolizine derivative for detecting target substances

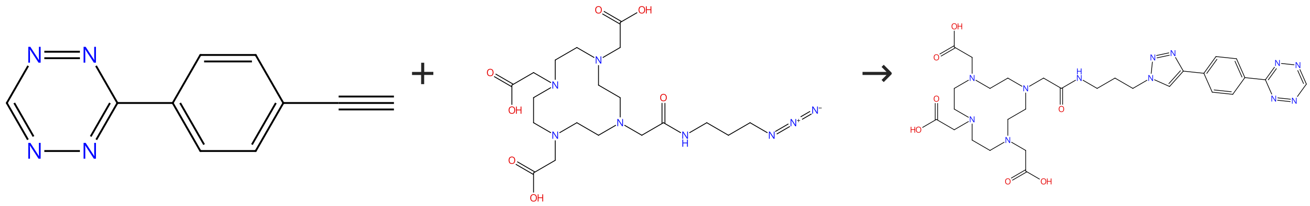
Assignee: Ajou University, Industry-Academic Cooperation Foundation

Korea, Republic of, KR2022063600 A 2022-05-17

PatentPak available

Scheme 48 (1 Reaction)

Steps: **1** Yield: **62%**



Suppliers (34)

Cu complex

31-614-CAS-31132831	Steps: **1** Yield: **62%**
1.1	**Reagents:** Sodium ascorbate, Copper sulfate **Catalysts:** 1,10-Phenanthrolinedisulfonic acid, 4,7-diphenyl-, sodium salt (1:2) **Solvents:** Dimethylformamide, Water; 1 - 3 h, rt

Experimental Protocols

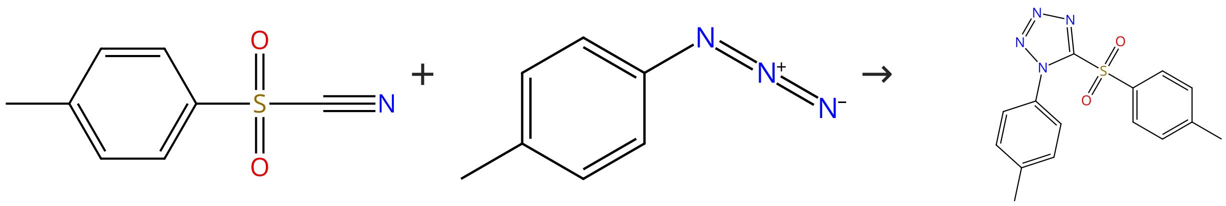
Lipophilicity and Click Reactivity Determine the Performance of Bioorthogonal Tetrazine Tools in Pretargeted In Vivo Chemistry

By: Steen, E. Johanna L.; et al

ACS Pharmacology & Translational Science (2021), 4(2), 824-833.

Scheme 49 (1 Reaction)

Steps: **1** Yield: **62%**



Suppliers (67)

Suppliers (31)

31-287-CAS-17367862	Steps: **1** Yield: **62%**
1.1	100 °C

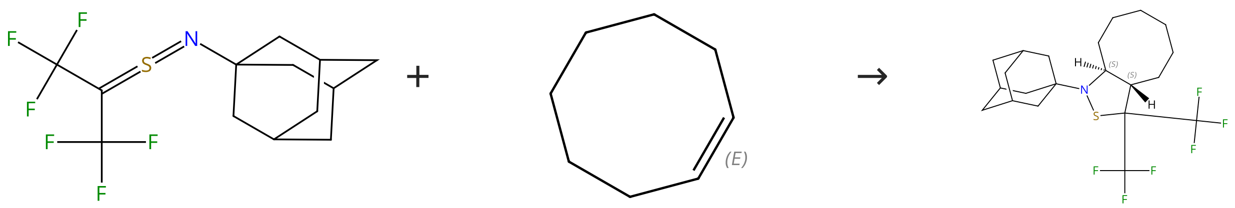
Phenyl azide

By: Pearson, William H.; et al

e-EROS Encyclopedia of Reagents for Organic Synthesis (2008), 1-9.

Scheme 50 (1 Reaction)

Steps: **1** Yield: **53%**



E/Z labels describe double bond geometry

Relative stereochemistry shown

Suppliers (3)

31-287-CAS-9843400	Steps: **1** Yield: **53%**
1.1	**Solvents:** Chloroform-*d*

Ambiguous reactivity of a fluorinated thiocarbonyl S-imide; unprecedented rearrangement under FVP conditions

By: Mloston, Grzegorz; et al

Tetrahedron (2000), 56(25), 4231-4238.